

ATTENTION UPS/Inverter manufacturers

**Precimec introduces
new range of products**

- Sine Wave HUPS Cards 650VA-4KVA
- Sine Wave Line Interactive UPS Cards 1KVA-3KVA
- Quasi Square Wave 3 in one Cards 650VA-1.8KVA
- High Frequency Transformerless HUPS Cards 400VA -1.2KVA
- Priority Solar Charge Controller 12V/24V /36V/48V in Capacities upto 40 Amps
- Solar Street Light Controller-for Single & Twin CFL
- SMPS Capacity 12V-48V DC upto 20Amps current
- Solar Inverter with RTC and Programmable Logic Control with LCD Display

Sample cards,
Technology and
Programmed IC
also available



Precimec Solar
charge controller



Precimec Hybrid
solar charge
controller



Precimec 650VA
Square wave



Precimec 300W
HF inverter



Precimec 1KVA Line
interactive sine wave

Precimec Products

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Q. What do the sensors (electrical) do? What are their types and applications?

Vikas
Bengaluru

A. In the field of measurement and instrumentation, the parameter to be measured (motion, pressure, temperature, etc) is first detected with the help of a sensor. The sensor converts the detected information into a suitable form (measurable currents and voltages) for acceptance in the later stages for decision making.

There are many types of sensors. To give you an idea of the basics of sensors, four types of sensors are discussed here.

Photoelectric sensor. These sensors have a light-emitting element that emits a visible or infrared beam. The beam interacts with the target and is received by a light-receiving element to determine the presence or type of the target.

There are two types of photoelectric sensors: reflective and through-beam. The reflective type is used to detect the

detectors contain a motion sensor that transforms the detection of movement into an electric signal. This can be achieved by measuring optical or acoustical changes in the field of view. The motion detector may be connected to a burglar alarm to alert the home owner or security service when any movement is detected.

Proximity sensors. Proximity sensors can detect metallic targets approaching them without requiring physical contact with the target. These are available in high-frequency oscillation (using electronic induction), magnetic (using a magnet) and capacitive types (using the change of capacitance).

Pressure sensors. These sensors use an internal pressure-sensitive element to convert any change in air pressure into an electrical signal.

Another related term that we come across is 'transducer.' A transducer is a device that converts one form of energy into another and vice versa. Electrical lamps and loudspeakers are very common examples of transducers. An electrical lamp converts

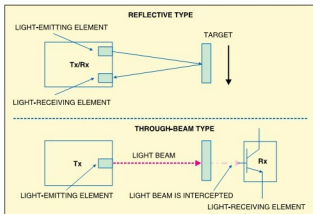
electrical energy into light energy, while a loudspeaker converts electrical energy into sound energy. Similarly, there are transducers that can convert other forms of energies.

Transducers find application in the fields of automobiles, manufacturing equipment and processes, aerospace, medicine

and robotics. Selection of a sensor depends on the type of parameter you want to detect and evaluate and also the purpose of your application.

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How photoelectric sensors work

light beam reflected off the target and the through-beam type is used to measure the change in light quantity caused by the target crossing the optical axis. Detection is based on the quantity of light received or the change in the quantity of reflected light. This method allows detection of such material as glass, metal, wood and liquid.

Motion detector. Electronic motion